comparison no trace of the entanglement could be detected, no faster than light-speed signal could be sent.

Nevertheless, even though entanglement can't be used for superluminal communication, one can't help feeling that long-distance correlations between particles are so bizarre that they've got to be useful for something extraordinary. In 1993, Bennett and his collaborators discovered one such possibility. They showed that quantum entanglement could be used for quantum teleportation. You might not be able to send a message at a speed greater than that of light, but if you'll settle for slower-than-light teleportation of a particle from here to there, entanglement's the ticket.

The reasoning behind this conclusion, while mathematically straightforward, is cunning and ingenious. Here's the flavor of how it goes.

Imagine I want to teleport a particular photon, one I'll call Photon A, from my home in New York to my friend Nicholas in London. For simplicity, let's see how I'd teleport the exact quantum state of the photon's spin—that is, how I'd ensure that Nicholas would acquire a photon whose probabilities of spinning one way or another were identical to Photon A's.

I can't just measure the spin of Photon A, call Nicholas, and have him manipulate a photon on his end so its spin matches my observation; the result I find would be affected by the observation I make, and so would not reflect the true state of Photon A before I looked. So what can I do? Well, according to Bennett and colleagues, the first step is to ensure that Nicholas and I each have one of two additional photons, let's call them Photons B and C, which are entangled. How we get these photons is not particularly important. Let's just assume that Nicholas and I are certain that even though we are on opposite sides of the Atlantic, if I were to measure Photon B's spin about any given axis, and he were to do the same for Photon C, we would find exactly the same result.

The next step, according to Bennett and coworkers, is *not* to directly measure Photon A—the photon I hope to teleport—since that turns out to be too drastic an intervention. Instead, I should measure a *joint* feature of